

Ng Guoyou

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Experience

SPH Media

SPH Media is a leading media organisation in Singapore with around 2500 employees.

Head, Digital Platform, Singapore

Apr 2024–Present

- Lead 15+ engineering reports across web application platforms, content infrastructure (licensing, serving, archival), and AI innovation labs, supporting core software development across multiple divisions serving millions of daily users.
- Shaped multi-year technical strategy and roadmaps for critical business functions including web platform architecture, content management, and archival systems.
- Drove SGD \$500K+ annual cost reduction through SaaS contract eliminations, cloud cost optimisations, and operational efficiency improvements across the engineering organisation.
- Built engineering foundation for scale through technical design review processes, development standards, and organisation-wide operational excellence practices (eg, structured logging, standardised alerting, on-call rotation).
- Successfully managed Singapore General Elections 2025 with zero downtime, ensuring platform reliability during the highest-traffic event of the year for the nation's leading news outlet.
- Managed AI innovation team that delivered production systems including newsroom research tools and semantic search capabilities.

Lead Software Engineer, Singapore

Apr 2022–Apr 2024

TypeScript, Node, Python, AWS, React, Vite, Docker, OpenSearch

- Led 10 developers across two teams to architect and deliver core platform services handling millions of daily page views across 15+ publications.
- Built custom web deployment platform for SSR/ISR Vite/React applications on AWS (Vercel-like internal solution), implementing edge middleware, incremental-static regeneration, Backend-for-Frontend framework, and automated deployment workflows—enabling rapid feature delivery while maintaining 99.9% uptime.
- Architected and delivered content infrastructure platform: OpenSearch-based content API (Algolia-like internal solution), ingestion pipelines processing 10K+ articles/day, image optimisation service using sharp (reducing bandwidth by ~40%), and content delivery APIs.
- Built analytics infrastructure including browser clickstream tracking script, backend ingestion service, and privacy-compliant session ID tracking—providing product and editorial teams with user behaviour insights.
- Led migration of 15+ publications from legacy Drupal/WordPress to a modern headless architecture, improving Core Web Vitals significantly and editorial workflow efficiency.
- Built proof-of-concept research projects including a production-scale liveblog and an edge-based ABAC policy engine.

Basis AI (acquired by Aicadium)

Basis AI was a startup that built a machine learning platform for data science teams.

Software Engineer, Singapore

Jan 2019–Oct 2020, Jun 2021–Apr 2022

TypeScript, Python, Rust, Webpack, React, Redux, Docker, Kubernetes, UX

- Employee #3. Full stack developer on a machine learning platform for training and deployment of data science workloads.
- Main contributor to frontend with Webpack, TypeScript, React+Redux. Created reusable React component library.
- Built client workload monitor in Rust; includes streaming ZIP and large file uploads for workload artifacts.
- Developed backend features in Python such as pagination and RBAC.
- Implemented Docker image caching for fast workloads using kaniko.
- Built company website in Next.js.
- Conducted basic UX research and user testing.

Credit Suisse

Software Engineer, Singapore

Nov 2020–Jun 2021

TypeScript, React, Redux, MobX, Webpack

- Maintained and implemented features for internal financial frontends related to securities order placement, disclosures, and compliance.

GovTech (Government Technology Agency)

GovTech is a Singapore government agency responsible for citizen-facing digital services.

Software Engineer, Singapore

Jan 2016–Jan 2019

TypeScript, React, Redux, Webpack, Python, Rust, Ruby on Rails

- [MyCareersFuture.gov.sg job portal](#), 2018. Orchestrated the codebase transition to TypeScript, Node, and React. Implemented features and bug fixes for the site.
- [OpenCerts.io](#), 2018. Created frontend tools to bulk-register academic transcripts on Ethereum using TypeScript and React.
- Data ingestion validator, API endpoint, and data visualisation application, 2017. Improved data ingestion and validation time by 50 times over existing Python script. Developed frontend application for visualising geographic data from scratch with IE11 support in Flow, React, and Deck.gl.
- [Business Grants Portal](#), 2016. Implemented features for and fixed bugs on forms with complex validation logic. Integrated authentication with SingPass SAML. Introduced linting for Ruby and JavaScript on a large codebase.

Roomorama

Intern, Software Engineer, Singapore

Summer 2013

Ruby on Rails

Education

Bachelor of Computing (Communications & Media), Hons.

National University of Singapore, 2011–2015

UX, HCI, and computer graphics focus

Personal

Singapore citizen. Served 2 years of National Service in the Singapore Armed Forces. Native English and Mandarin Chinese speaker.

Personal Projects

- [A now-playing desktop widget for Windows using Rust, Tauri, and Svelte](#), 2023
- [A zero-config, static, file-based album generator](#), 2022
- [TypeScript library to fetch file listings and individual files from a remote ZIP file](#), 2021
- [Browser extension for organising downloads with >3000 users](#), 2018
- [Retro image filtering application in React and Redux](#), 2017
- [Chatbot for IRC, Slack, Discord in Ruby with a plugin system](#), 2016
- [Toy audio synthesiser library in Rust with MIDI support and WebAssembly interface](#), 2014
- [Sharing sessions and presentations](#)

More projects on [GitHub](#).